An inductive-recursive approach to ordinals

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Abstract

Rather than using ordinal notations, we introduce a inductive-inductive-recursive definition of ordinals.

Each ordinal a can be uniquely written as the sum $\omega^{a_0} + \ldots + \omega^{a_n}$ where $a_0 \geq \cdots \geq a_n$ are ordinals, and hence can be equivalently represented by the decreasing sequence $a_0, \ldots, a_n$. Because the elements of such a decreasing sequence are also decreasing sequences, ordinals (*i.e.* decreasing sequences) have to be defined simultaneously with an ordering on them. Moreover, when inductively constructing a new decreasing sequence, a head function for such sequences seems needed to guarantee that the inserted head is greater than (the head of) the tail. Therefore, we have the following *inductive-inductive-recursive* definition of ordinals.

Definition 1. We mutually define a type $\mathcal{O}$, a relation $<$ over $\mathcal{O}$ and a function $\mathrm{fst} : \mathcal{O} \to \mathcal{O}$ as follows

- The type $\mathcal{O}$ is defined with two cases

  ($\mathcal{O}1$) $0 : \mathcal{O}$,

  ($\mathcal{O}2$) if $a, b : \mathcal{O}$ with $a \geq \mathrm{fst}(b)$, then $(a, b) : \mathcal{O}$.
  If a, b denotes some ordinals, then (a, b) denotes the ordinal $\omega^a + b$. Hence we usually write the latter instead of pairs.

- The relation $<$ is given by the following axiom and rules

  (<1) If $a \neq 0$ then $0 < a$.

  (<2) If $a < b$ then $\omega^a + c < \omega^b + d$
  for any $c, d : \mathcal{O}$ with $a \geq \mathrm{fst}(c)$ and $b \geq \mathrm{fst}(d)$.

  (<3) If $b < c$ then $\omega^a + b < \omega^a + c$
  for any $a : \mathcal{O}$ with $a \geq \mathrm{fst}(b)$ and $a \geq \mathrm{fst}(c)$.

- The function $\mathrm{fst} : \mathcal{O} \to \mathcal{O}$ is defined by cases on its argument

$$\begin{aligned} \mathrm{fst}(0) &:\equiv 0 \\ \mathrm{fst}(\omega^a + b) &:\equiv a. \end{aligned}$$

As mentioned above, an element of type $\mathcal{O}$ is a decreasing sequence of elements of type $\mathcal{O}$. Specifically, ($\mathcal{O}1$) defines the empty sequence, and ($\mathcal{O}2$) forms a new sequence (a, b) whenever a is greater than or equal to the head of b. The empty sequence denotes the ordinal 0, while the nonempty one (a, b) denotes $\omega^a + b$. When b is 0, we omit it and simply write ω^a.

Lemma 2. The relation $<$ is a strict total order on $\mathcal{O}$.

Proof. Irreflexivity and transitivity are proved by structural induction on the relation. Trichotomy is proved by induction on the arguments. □

Because of the mutual definition, one has to simultaneously define addition of ordinals and prove Lemma 4 of a property of addition.

Definition 3. Addition of ordinals is defined as follows

$$\begin{array}{rcl} 0 + b & :\equiv & b \\ a + 0 & :\equiv & a \\ (\omega^a + c) + (\omega^b + d) & :\equiv & \begin{cases} \omega^b + d & \text{if } a < b \\ \omega^a + (c + (\omega^b + d)) & \text{otherwise} \end{cases} \end{array}$$

where the last case is well-defined because $a \geq \mathrm{fst}(c + (\omega^b + d))$ by Lemma 4.

Lemma 4. For any $a, b, c : \mathcal{O}$, if $a \geq \mathrm{fst}(b)$ and $a \geq \mathrm{fst}(c)$ then $a \geq \mathrm{fst}(b + c)$.

Proof. By induction on b and c. It's trivial when $b \equiv 0$ or $c \equiv 0$.

Let $b \equiv \omega^d + u$ and $c \equiv \omega^e + v$. (1) If $d < e$, then

$$\mathrm{fst}(b + c) = \mathrm{fst}((\omega^d + u) + (\omega^e + v)) = \mathrm{fst}(\omega^e + v)) = e$$

by the definition, and we already have $a \geq e$ by assumption. (2) If $d \geq e$, then

$$\mathrm{fst}(b + c) = \mathrm{fst}((\omega^d + u) + (\omega^e + v)) = \mathrm{fst}(\omega^d + (u + (\omega^e + v))) = d$$

by the definition and the induction hypothesis, and we also have $a \geq d$ by assumption. □